

Question	Marking Guidance	Mark	Comments
05.1	$ \begin{array}{c} \text{R} \\ \\ \text{H}_2\text{N} \text{ — } \text{C} \text{ — } \text{COOH} \\ \\ \text{H} \end{array} $	1	Accept other correct representations.
05.2	- More than one codon codes for a single amino acid; - Suitable example selected from Table 1;	2	- Accept 'triplet' or 'sequence of 3 bases/nucleotides' for 'codon'. - Reject 'production/produces' for 'codes for'. Do not infer mp1 from mp2.
05.3	1395;	1	Accept 1398 and 1401 (for those that include start and/or stop codons) Allow 2796 or 2802 or 2790 Ignore 'bases/base pairs/bp/bps' written after the numerical answer.
05.4	☒ CAA → CGA	1	

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05.5	1. (Both) negatively charged to positively charged change in amino acid; 2. Change at amino acid 300 does not change the shape of the <u>active site</u> OR Change at amino acid 300 does not change the tertiary structure OR Change at amino acid 300 results in a similar tertiary structure; 3. Amino acid 279 may have been involved in a (ionic, disulfide or hydrogen) bond and so the shape of the active site changes OR Amino acid 279 may have been involved in a (ionic, disulfide or hydrogen) bond and so the tertiary structure changed; OR Amino acid 279 may be in the active site and be required for binding the substrate;	3	2. and 3. Reference to 'shape' of active site only needed once. 3. Both parts are required for each mark option. 3. For 'a bond' reject peptide bond.

